
UNIT 15 MOBILE APPLICATION LANGUAGES

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15.0 INTRODUCTION

Mobile applications can be developed in various languages such as Java Python and others. in this chapter we will look at some of them salient features of this programming languages and sample code for developing mobile applications.

15.1 OBJECTIVES

After going through this unit, you should be able to

- understand key concepts of XML,
- understand various types of device markups,
- understand the details of WML,
- understand HTML5 standard

15.2 INTRODUCTION TO J2EE

Java is one of the most popular programming languages. Java helps us to depict Real world objects in the programming language. application developer uses the libraries provided by the software development kit additional him the programmer can also use various third party libraries and APIs to integrate with external interfaces.

There are various integrated development environment such as Eclipse, IntelliJ , Jdeveloper, Microsoft Visual Studio that helps developers to develop a java based applications.

The standard edition of Java is mainly used for developing the libraries and client applications. the Enterprise addition of Java (J2EE) is mainly used for developing web applications, server applications and Enterprise applications. J2EE handles the distributed applications spread across

15.2.1 Brief Overview of Java

Java is an object oriented language. the model with real world entities as Java objects we depict the state of the object using the variables. we define the behavior of the object using the methods. Java is a interpreted language and hence it runs on any operating system and any architecture (such as x86 or ARM). The JAVA SDK provides many popular libraries and API for the development. for instance the package Java total and provides the core datatypes and main classes.

The Java code is compiled into bytecodes. The Java Virtual Machine (JVM) runs the byte code on various platforms such as Linux, Windows and others.

The package java.math provides the mathematical functions. The package java.net provides the leather is for network connections; the package java.io provides input output and read write operations; the package Java dot security provides various classes for encryption decryption and other security related concerns.

Given below organ main constructs of Java:

- Class: A class is a logical unit that represents a real-world entity. A class serves as a template and groups the properties and the behavior of a real world entity in a class. The properties are depicted as fields in the class and the behavior is defined using the methods of the class. For instance a person class consists of fields such as name and phone number and behavior such as updating the phone number. An instance of the class is called an object. Object essentially encapsulates the state of the real world entity. For instance we can assign the name as Kumar and phone number as 1 2 3 1 2 3 1 2 3 4 for the person object.
- Method: A method is a function that depicts the behavior of the class.
- Bean: A bean encapsulates a key value pair in a class. We use beans to represent enmities such as database table or an entity of an XML document.
- Relationship between classes: There are various kinds of relationship between classes. A child class can inherit the values and behavior from its parent class. A class can also be composed of another class.
- Interface: The interface specified the behavior and service as a contract for implementer.
- Threads: Thread is a lightweight runnable object. We span multiple threads to process in parallel.
- Servlets: The servlets are web components that are used to render the web pages. The servlets have a lifecycle of its own.

15.2.2 N-tier Architecture in J2EE

In this section we discuss the key elements of n-tier J2EE architecture. Figure 1 depicts various tiers in a J2EE enterprise application. The presentation tier provides components such as responsive design, personalization etc. The services tier provide various services for workflow, multi-lingual services and others. The

business tier consists of business components such as blogs, search, email etc. We can have additional components related to integration, enterprise systems etc.

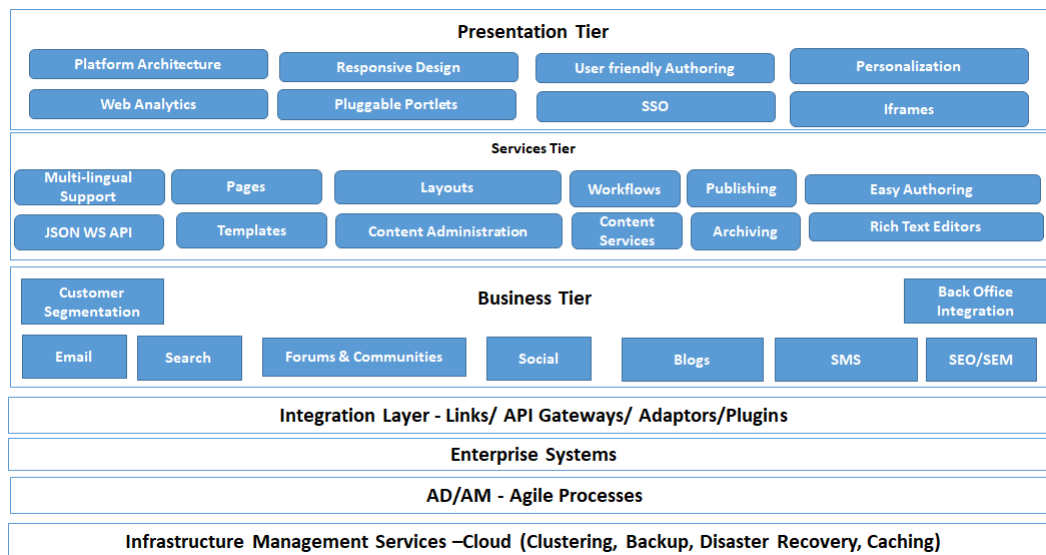


Figure 1 Sample N-Tier Architecture

We have depicted the systems view of the n-tier architecture in figure 2. Systems of interaction include the presentation components such as UI applications, analytics, caching, pages etc. Systems of differentiation include the web frameworks and core platform. Systems of integration consist of components such as API gateway, ESB and other middleware components. We have systems of record that store single source of truth for enterprise data. Systems of record include ERPs, enterprise databases and such.

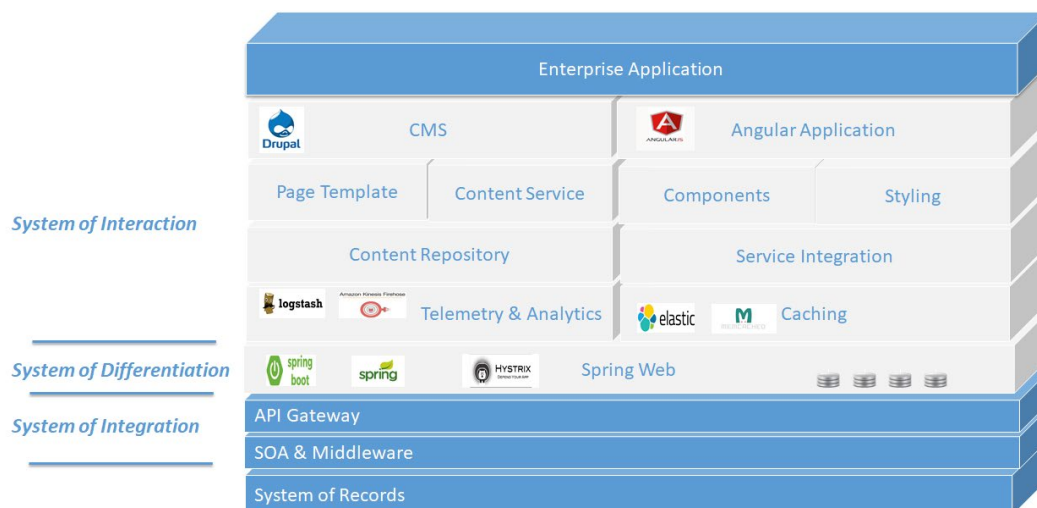


Figure 2 Systems View of the architecture

We have depicted various components involved in processing the request pipeline of the J2EE application. The request is initially handled by the cache service that provide edge-level caching. In the next step Okta-based authentication module authenticates the user. Post successful authentication, one of the functional modules (such as billing, payment, account etc.) handles the request and will leverage the components in the presentation layer.

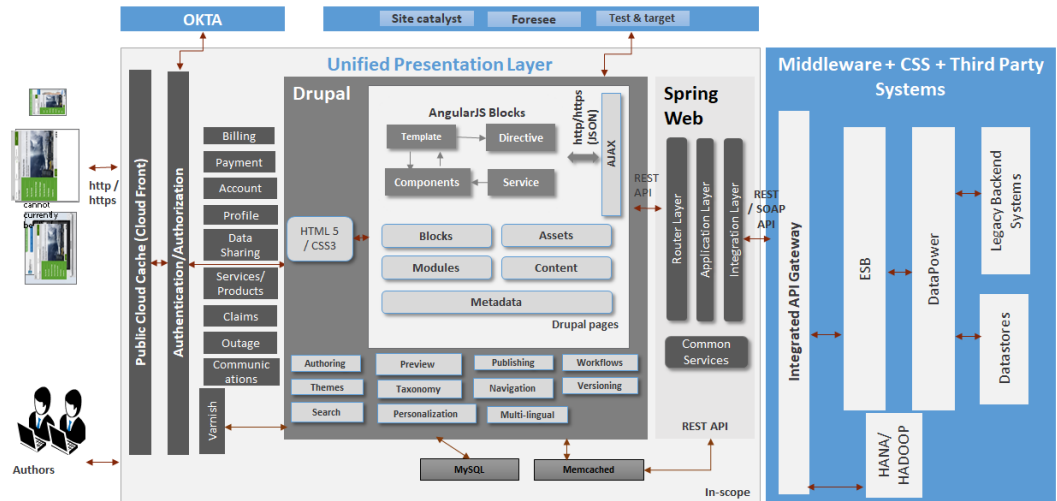


Figure 3 Request pipeline processing in enterprise application

The presentation layer modules such as HTML 5, CSS3 modules, web pages, search provide the end user experience components. The Spring web layer provides request routing, common services and REST interfaces. The service requests are then handled by the middleware systems such as ESB which retrieves data from the enterprise systems.

15.2.3 Patterns and design considerations for J2EE

Let us look at the main design considerations and advantages of J2EE.

Key advantages of J2EE

The main motivating factors/advantages of J2EE platform are as follows:

- J2EE technologies support separation of concerns by providing components and frameworks for each layer. It also separates business concerns from system concerns by providing container level services like resource management, lifecycle management. This allows the developers to focus mainly on business logic and enable to bring the solution faster to market.
- J2EE technologies support various standards such as JDBC, JSP, XML, SOAP, JMS, JNDI, Portlet, JTA and such.
- J2EE technologies support heterogeneous software and hardware avoiding vendor lock-in for the organization. It also future-proof technology roadmap as it keeps pace with technology advancements
- J2EE technologies provide robust mechanisms for session management, transaction management, security features and would provide required level of scalability and performance when configured on appropriate hardware
- J2EE technologies offer various integration mechanisms ranging from service based integration to message based integration to API based integration. This would help the organizations to easily integrate with external systems as well as in-house legacy applications.

Key design considerations for the enterprise solution

Following are the key design considerations while designing the solution:

- **Extensibility:** The solution should allow easy addition/extension of new functionality.
- **Modularity:** The solution should provide intra-layer abstraction by allowing the individual layer components to be independently modified with minimal impact on components in other layers.
- **Scalable:** The solution should satisfy the scalability requirements explicitly stated.
- **Secure:** The solution should provide security features at all levels to protect data and transaction integrity.
- **Open standards:** During development of integration interfaces and other components, open standards would be followed to prevent vendor lock-in.

The n-tier J2EE application typically uses Model View Controller (MVC) architectural pattern for a layered architecture. We have listed other common design patterns used in the enterprise application.

Design Pattern	Key design pattern	Brief Details
Presentation Layer		
	View Helper	View Helper Pattern in presentation layer for creating a custom Tag required for the solution
	Composite View	Composite View pattern in presentation layer for including JSF fragments for each page. The fragments are related to header, footer and other navigation elements to provide consistent look and feel/brand identity
	Front controller	Front controller is the single interface that handles all the web requests
Business Layer	Business Delegate	Business Delegate pattern to abstract the business service from presentation layer
	Session Facade	Session Façade Pattern to abstract the business components from the clients
	Dependency Injection/Inversion of control	Provides declarative way for defining the dependencies
Integration Layer	Web service proxy	Web service proxy pattern to invoke the internal web services like pricing system service and inventory system service to decouple the client and

		actual service details
	Business Object	Business objects encapsulate the data and operations of business entity
	Abstract Factory	The abstract factory defines the interfaces for constructing a family of related classes
	Data Access Object	The data access object (DAO) abstracts the database operations
	Service Activator for JMS	Service Activator Pattern to asynchronously receive any messages from the messaging system
Core Java Patterns		
	Singleton	Provides a single instance of the class. Mainly used to instantiate the utility class.

15.3 INTRODUCTION TO J2ME

Java 2 Micro Edition (J2ME) is the set of Java APIs that can run application on mobile devices with limited memory and resources.

The main components of J2ME are configurations and profiles. The configuration specify the minimally-required Java classes and Java Virtual Machine features required for a specific device. The configuration groups the devices with similar resource. The profile provides APIs leveraging the configuration to provide the end device's run time environment. Mobile information device profile and personal profile are two main kinds of profile used by J2ME devices. Information module profile is used for devices like vending machines, embedded systems which have no display or minimal display.

Mobile information device profile is used mainly in the mobile game development.

15.3.1 Sample Java program for file parsing

Modern applications heavily use JSON (JavaScript Object Notation) for data exchange. In a n-tier mobile application, the Java-based mobile app needs to parse and write the JSON file. In this section we discuss the sample Java code for achieving this.

Let us consider the sample JSON of student's data as given below:

```
{  
  "name": "Kumar",  
  "rollno": "1234",  
  "course": "BA"  
}
```

Given below is the sample Java code to create the JSON file in above format. In this example we have used the json library.

```
import java.io.FileNotFoundException;
import java.io.PrintWriter;
import java.util.LinkedHashMap;
import java.util.Map;
import org.json.simple.JSONArray;
import org.json.simple.JSONObject;

public class SampleJSONCreator
{
    public static void main(String[] args) throws FileNotFoundException
    {
        // Create the JSONObject
        JSONObject jo = new JSONObject();

        // populate the JSON object with the sample data
        jo.put("name", "Kumar");
        jo.put("rollno", "1234");
        jo.put("course", "BA");

        // create the StudentData JSON file
        PrintWriter pw = new PrintWriter("StudentData.json");
        pw.write(jo.toJSONString());

        pw.flush();
        pw.close();
    }
}
```

Given below is the sample Java code to parse the JSON string

```
import org.json.JSONArray;
import org.json.JSONObject;

public class SampleJSONParser{

    public static void main(String[] args) {

        String jsonString = "{"
            + " \"name\": \"kumar\","
            + " \"rollno\": \"1234\","
            + " \"course\": \"BA\"
            + "}";

        //Create the JSONObject based on the jsonString
        JSONObject studentJSON = new JSONObject(jsonString);

        //Read the JSON value
        String name = studentJSON.getString("name");
        System.out.println("Student Name: "+name+"\n");
    }
}
```

}

☞ Check Your Progress 1

1. ____ edition of Java is mainly used for developing the libraries and client applications
2. ____ addition of Java is mainly used for developing web applications
3. A ____ is a logical unit that represents a real-world entity
4. ____ encapsulates a key value pair
5. ____ provide intra-layer abstraction by allowing the individual layer components to be independently modified

15.4 INTRODUCTION TO ANDROID

Android is an open source mobile development platform. Android is based on Java programming language. Android has a huge platform ecosystem and is widely used across various mobile devices today.

Android supports various connectivity such as GSM, Bluetooth, Wi-Fi , LTE, CDMA etc. Android supports lightweight SQLite database. We can run an chrome web browser on Android. Android supports various other features such as multi touch multitasking widgets multi language etc. Android is hugely popular and it has the largest supported mobile device.

Android operating system provides many inbuilt applications such as camera alarm, calculator, contacts, email, calendar, media player, albums, clock etc. Android supports various libraries such as SQLite, SSL etc. Android runs on Linux kernel.

We can use in-built SMS Manger API for sending the SMS. A sample code snippet is shown below:

```
SmsManager = SmsManager.getDefault();  
smsManager.sendTextMessage("phoneNo", null, "Test Message", null, null);
```

In Android we can use intent to pass the data from one component to another or to the external interface.

Let us look at two examples of using Intent for making a phone call and for sending out email.

Given below is the code snippet that uses the intent for making a call.

```
public class MainActivity extends AppCompatActivity {  
    private Button;  
  
    @Override  
    protected void onCreate(Bundle savedInstanceState) {  
        .
```

```

        .
        .

button.setOnClickListener(new View.OnClickListener() {
    public void onClick(View arg0) {
        Intent callAction = new Intent(Intent.ACTION_CALL);
        callAction.setData(Uri.parse("tel:1231231234"));

        startActivity(callAction);
    }
});
}
}

```

Given below is another Android example code snippet using Intent for sending out email.

```

protected void doEmail() {

    String[] reciever = {"receiver@example.com"};
    String[] copyemail = {"supervisor@examle.com"};
    Intent emailAction = new Intent(Intent.ACTION_SEND);

    emailAction.setData(Uri.parse("mailto:"));
    emailAction.setType("text/plain");
    emailAction.putExtra(Intent.EXTRA_EMAIL, reciever);
    emailAction.putExtra(Intent.EXTRA_CC, copyemail);
    emailAction.putExtra(Intent.EXTRA_SUBJECT, "Test Message");
    emailAction.putExtra(Intent.EXTRA_TEXT, "This is a test message");

    try {
        startActivity(Intent.createChooser(emailAction, "Sending email."));
        finish();
    } catch (android.content.ActivityNotFoundException ex) {

    }
}

```

15.5 PYTHON AND OTHER LANGUAGES

Python is a scripting language and is one of the most popular open source languages. Python is widely used in the platforms big data applications and machine learning platforms. Python provides many libraries and built-in functions for processing and data networking. Python is portable across various OS platforms.

Python supports large set of libraries including database, multimedia ,networking, graphical user interface, image processing, machine learning mobile apps, text processing, automation and so on. Python is widely used in machine learning projects using library such as keras,tensorflow, pytorch.

Let us look at an example for sending the SMS message to a mobile using Python. We can leverage various libraries such as twilio to send the SMS. Given below is the sample code snippet for sending the SMS using twilio library in Python:

```
from twilio.rest import Client
client = Client(sid, auth_token)

message = client.messages \
    .create(
        body='Sample SMS',
        from_ = 1231234123,
        to = 2451231234
    )

print(message.sid)
```

15.5.1 Swift

Swift is an open source intuitive compiled programming language designed by Apple. Swift is used in Apple iOS, devices such as Apple watch, Mac and others. Swift language uses objective C library. The Swift manages the memory automatically.

Swift provides native error handling and provides other features such as flexible enumerations, closure syntax, structs and classes.

Let us look at sample code for handling key functions in Swift

Given below is a code snippet for making a phone call in Swift -

```
func makeCall(phoneNo: String) -> Bool {
    if let pNo = URL(string: "tel://" + phoneNo) {
        if UIApplication.shared.canOpenURL(pNo) {
            UIApplication.shared.openURL(pNo, options: [:], completionHandler:
nil)
            return true
        }
    }

    return false
}
```

For sending SMS we can leverage MFMessageComposeViewController and for sending email we can leverage MFMailComposeViewController

15.5.2 Microsoft Dot Net Framework

Microsoft.net provides various components for graphical user interface data processing data validation and SQL. languages such as C#, Visual Basic, C++ are part of the studio .NET framework. microsoft.net provides various features such

as multilingual support XML processing common language runtime and others. It provides an extensive library for database processing, Cryptography XML processing data binding web frameworks and others.

Check Your Progress 2

1. ____ can be used for sending email in Android
2. In Android we can use ____ to pass the data from one component to another
3. __ is used for developing Apple iOS applications
4. In Swift we use _____ for sending SMS.

15.6 SUMMARY

In this unit, we started discussing the main constructs of the standard edition of the Java programming language. The standard edition of Java is used for developing libraries and client applications and enterprise edition is used for developing web application and enterprise applications. For developing the enterprise applications we use n-tiered design with model view controller pattern where we create separate components in each tier handling distinct responsibility. Java micro edition is used for developing mobile gaming applications. Android is one of the most popular platforms. Android supports various connectivity such as GSM, Bluetooth, Wi-Fi , LTE, CDMA . In Android we can use intent to pass the data from one component to another or to the external interface. Using Intent we can send SMS or email within Android. Python is one of the most popular open source scripting languages. In Python we can use libraries for implementing data processing functions, Machine learning operations and for sending email, SMS.

15.7 SOLUTIONS/ANSWERS

Check Your Progress 1

1. Standard
2. Enterprise
3. Class
4. Bean
5. Modularity

Check Your Progress 2

1. SMSManager
2. Intent
3. Swift
4. MFMessageComposeViewController

15.8 FURTHER READINGS

References

Mobile Computing 3rd Edition by Raj Kamal -

<https://www.amazon.in/Mobile-Computing-Raj-Kamal/dp/0199455414>